

Section 2 – Language, Data and Knowledge

LLM with data and knowledge integration

COMM116 Generative AI Applications

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University
of Exeter

Agenda

- What is data? What is knowledge?
- The lifecycle of knowledge in LLMs
- Multimodality – using medical imaging as an example
- Project discussion

Language Models: Predicting the probability of next words



what is the |
what is the **weather**
what is the **meaning of life**
what is the **dark web**
what is the **xfl**
what is the **doomsday clock**
what is the **weather today**
what is the **keto diet**
what is the **american dream**
what is the **speed of light**
what is the **bill of rights**

Google Search

I'm Feeling Lucky



Language models

2.1 Unidirectional Language Models

Given an input sequence of tokens $\mathbf{w} = [w_1, w_2, \dots, w_N]$, unidirectional language models commonly assign a probability $p(\mathbf{w})$ to the sequence by factorizing it as follows

$$p(\mathbf{w}) = \prod_t p(w_t | w_{t-1}, \dots, w_1). \quad (1)$$

$$p(w_t | w_{t-1}, \dots, w_1) = \text{softmax}(\mathbf{W}\mathbf{h}_t + \mathbf{b}) \quad (2)$$

where $\mathbf{h}_t \in \mathbb{R}^k$ is the output vector of a neural network at position t and $\mathbf{W} \in \mathbb{R}^{|\mathcal{V}| \times k}$ is a learned parameter matrix that maps $\mathbf{h}_t$ to unnormalized scores for every word in the vocabulary $\mathcal{V}$.

Petroni F, Rocktäschel T, Riedel S, Lewis P, Bakhtin A, Wu Y, Miller A. *Language Models as Knowledge Bases?*. In EMNLP-IJCNLP 2019 Nov (pp. 2463-2473).

ELMo

Slides by ELMo's authors from <https://www.slideshare.net/shuntaroy/a-review-of-deep-contextualized-word-representations-peters-2018>

Right figure from 1. Manning CD, Clark K, Hewitt J, Khandelwal U, Levy O. Emergent linguistic structure in artificial neural networks trained by self-supervision. PNAS. 2020 Jun 3.

BERT

2.2 Bidirectional “Language Models”²

given an input sequence $\mathbf{w} = [w_1, w_2, \dots, w_N]$ and a position $1 \leq i \leq N$, we want to estimate $p(w_i) = p(w_i | w_1, \dots, w_{i-1}, w_{i+1}, \dots, w_N)$ using the left and right context of that word.

► biLMs consist of forward and backward LMs

◆ Forward: $p(t_1, t_2, \dots, t_N) = \prod_{k=1}^N p(t_k | t_1, t_2, \dots, t_{k-1})$

◆ Backward: $p(t_1, t_2, \dots, t_N) = \prod_{k=1}^N p(t_k | t_{k+1}, t_{k+2}, \dots, t_N)$

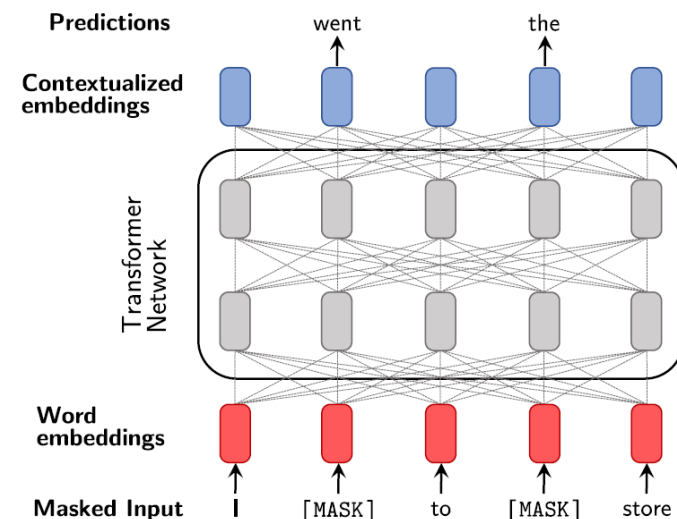


Fig. 3. A high-level illustration of BERT. Words in the input sequence are randomly masked out and then all words are embedded as vectors in $\mathbb{R}^d$. A Transformer network applies multiple layers of multiheaded attention to the representations. The final representations are used to predict the identities of the masked-out input words.

(Neural) Language Models (LM)

Unidirectional language model: decoder-only

$$p(\mathbf{w}) = \prod_t p(w_t | w_{t-1}, \dots, w_1).$$

$$p(w_t | w_{t-1}, \dots, w_1) = \text{softmax}(\mathbf{W}\mathbf{h}_t + \mathbf{b})$$

$\mathbf{h}_t$ is from a neural network (e.g., Transformer, in 2017)

Elmo (in 2018)

► biLMs consist of forward and backward LMs

◆ Forward: $p(t_1, t_2, \dots, t_N) = \prod_{k=1}^N p(t_k | t_1, t_2, \dots, t_{k-1})$

◆ Backward: $p(t_1, t_2, \dots, t_N) = \prod_{k=1}^N p(t_k | t_{k+1}, t_{k+2}, \dots, t_N)$

BERT (in 2018-2019)

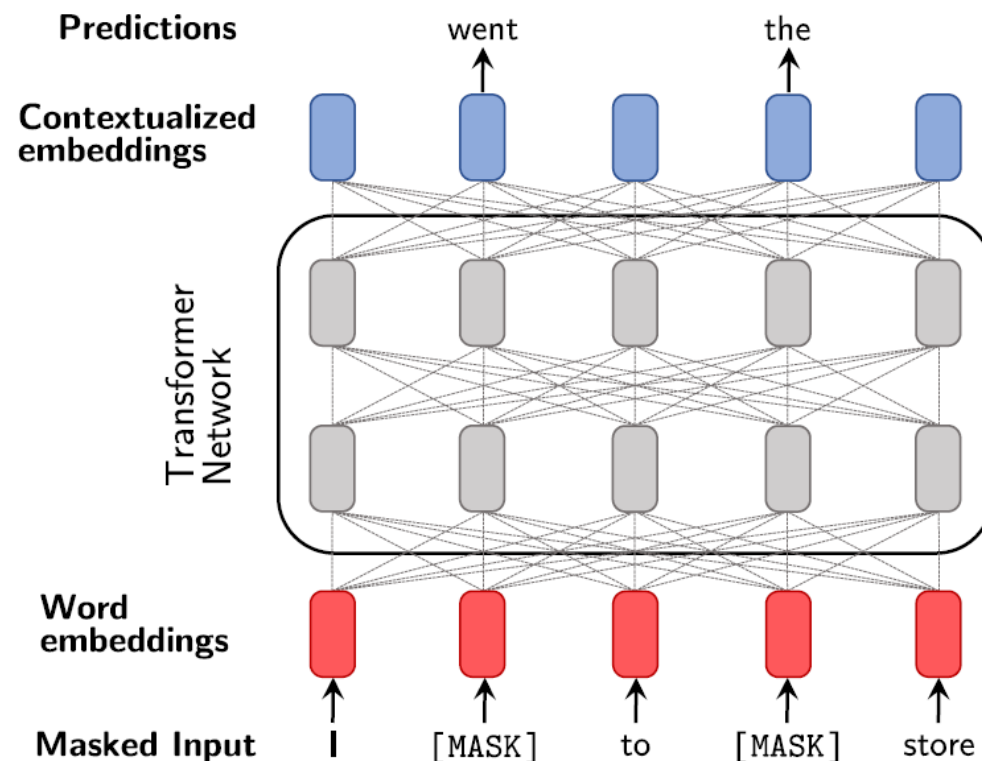
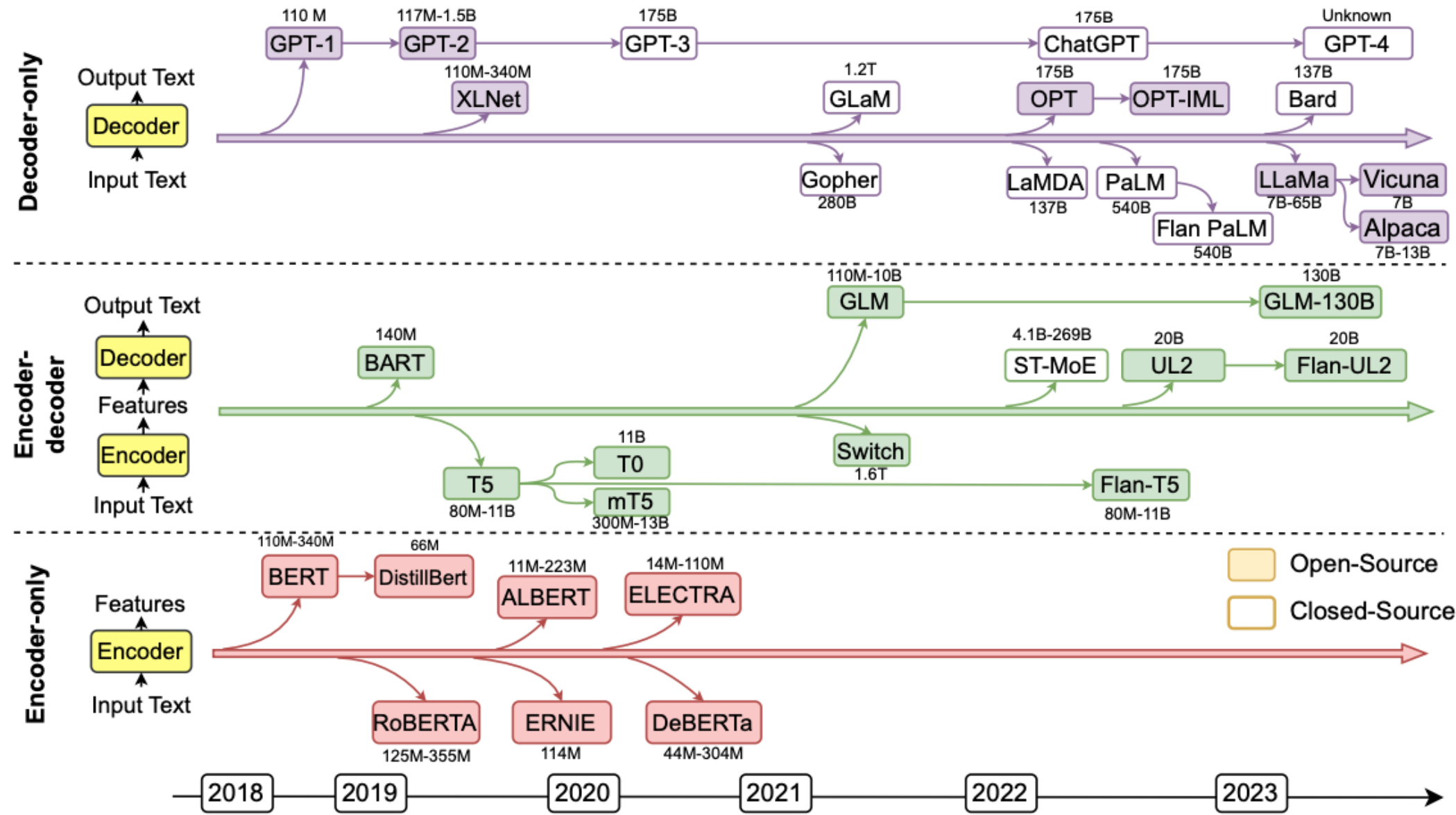


Fig. 3. A high-level illustration of BERT. Words in the input sequence are randomly masked out and then all words are embedded as vectors in $\mathbb{R}^d$. A Transformer network applies multiple layers of multiheaded attention to the representations. The final representations are used to predict the identities of the masked-out input words.

Language models till mid 2023



Fundamental issues of LLMs

- Hallucination
- Opaqueness
- Staleness
- Incompleteness

(The above from Hogan et al., 2025)

- Ethical concerns: harmful, trustworthiness, uncertainty, etc. – We will look at some of them next week

Hogan, A., Dong, X. L., Vrandečić, D., & Weikum, G. (2025). Large Language Models, Knowledge Graphs and Search Engines: A Crossroads for Answering Users' Questions. *arXiv preprint arXiv:2501.06699*.

Discussion

- What is knowledge?
- Does LLM have knowledge?

What is knowledge?

- We understand by "knowledge" all kinds of facts about the world.
 - Commonsense knowledge, domain knowledge, etc.
- Knowledge representation: Representation refers to a symbol or thing which represents ('refers to', 'stands for') something else.
- Knowledge base
- Does LLM have knowledge?

Knowledge base

- A knowledge base is a set of *sentences*. Each sentence is expressed in a language and represents some assertions of the world. (Russel, S. & Norvig, P., Section 7.1 Knowledge-Based Agents, in *Artificial Intelligence: A Modern Approach*, Fourth Edition, Global Edition, 2022, p. 227)

dbpedia.org/page/University_of_Exeter

DBpedia Browse using Formats Faceted Browser Sparql Endpoint

About: [University of Exeter](#)

An Entity of Type: [Public university](#), from Named Graph: <http://dbpedia.org>, within Data Space: [dbpedia.org](#)

Public research university located in Exeter, United Kingdom



Property	Value
dbo:affiliation	dbr:European_University_Association dbr:Association_of_Commonwealth_Universities dbr:Association_of_MBAs dbr:Russell_Group dbr:Universities_UK dbr:Worldwide_Universities_Network dbr:GW4 dbr:SETsquared
dbo:campus	dbr:Penryn,_Cornwall dbr:St_Luke's_Campus dbr:Streatham_Campus
dbo:chancellor	dbr:Michael_Barber_(educationist)
dbo:city	dbr:Devon dbr:Exeter dbr:Penryn,_Cornwall

Knowledge representation in search engine

- "Knowledge panel" - KG

<https://support.google.com/knowledgepanel/answer/9163198>

VS.

- "AI overviews" – LLM/GenAI

<https://search.google/ways-to-search/ai-overviews/>

The screenshot shows a Google search for "university of exeter". The search bar is at the top with the Google logo. Below the search bar, there are tabs for "AI Mode", "All", "Images", "News", "Maps", "Videos", "Shopping", "More", and "Tools". The "All" tab is selected.

The search results for "University of Exeter" are displayed. The top result is the official website, "University of Exeter | A Top Worldwide University", with a URL of "https://www.exeter.ac.uk". Below the title, there is a description: "A world-leading UK university, part of the prestigious Russell Group, offering outstanding research and teaching across three campuses in the South West."

Below the description, there are several sections with expandable content:

- TimeCAP: A Channel-Aware ...** (TimeCAP: A Channel-Aware Pre-Training Framework for ...)
- Term dates** (2025-26 term dates · Autumn term · Spring term · Summer term ...)
- Undergraduate study** (Explore studying as an undergraduate at the University ...)
- Postgraduate study** (Explore studying as a postgraduate at the University of ...)
- Undergraduate Degrees** (Find your perfect undergraduate degree at the University of ...)
- Our campuses** (The University of Exeter is spread over three campuses, two in ...)

Below these sections, there is a link to "More results from exeter.ac.uk »".

At the bottom of the search results, there is a section titled "People also ask" with a question: "Is Exeter University prestigious?".

On the right side of the search results, there is a "Knowledge panel" for "University of Exeter". It includes a 4.3-star rating from 500 Google reviews, the location "University in Exeter, England", and buttons for "Website", "Directions", "Reviews", "Save", "Share", and "Call". Below the buttons, there is a description: "Main campus of 1950s-founded university, with landscaped hilltop grounds including a sculpture walk." It also lists the address "Stocker Rd, Exeter EX4 4PY", the founding year "1851", and tuition fees: "Undergraduate tuition and fees: 9,000 GBP, International tuition 17,500 GBP (2015 – 16)". Other information includes "Total enrollment: 22,540 (2017)", colors "White, Green", and campus details: "Campus: Streatham – 350 acres (140 ha); Penryn – 100 acres (40 ha); St Luke's – 16 acres (6.5 ha)". The number of undergraduates is listed as "23,865 (2023/24)".

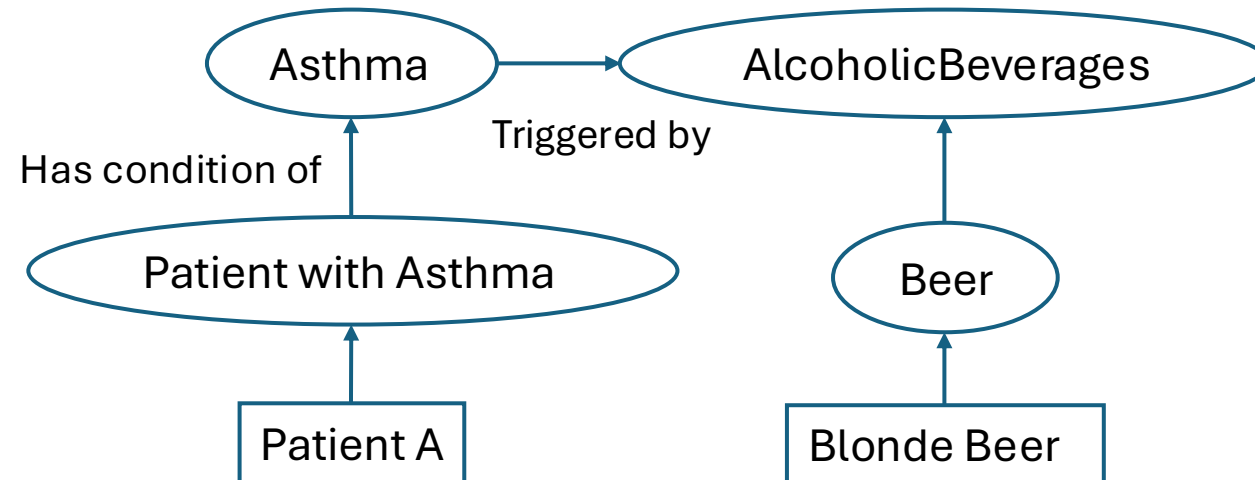
Google's Knowledge Graph

- Amit Singhal. Introducing the Knowledge Graph: things, not strings. May 16, 2012.
<https://blog.google/products-and-platforms/products/search/introducing-knowledge-graph-things-not/>
- Google. Introducing the Knowledge Graph.
<https://youtu.be/mmQl6VGvX-c>. May 16, 2012/.
- Google. Knowledge panels. How Google's Knowledge Graph works. Accessed Jan 29, 2026.
<https://support.google.com/knowledgepanel/answer/9787176?hl=en>



Knowledge Graphs (KG)

- Knowledge graphs: **entities** in the domain of interest & the **relationships** between these entities. (Hogan et al., 2021, kgbook.org)
- KG provides **reasoning**, or predictable inference.



International Classification of Diseases, Version 10

Last uploaded: November 18, 2021

SummaryClassesPropertiesNotesMappingsWidgets

Jump to:

Certain conditions originating in the perinatal period

Certain infectious and parasitic diseases

Arthropod-borne viral fevers and viral haemorrhagic diseases

Bacterial, viral and other infectious agents

Certain zoonotic bacterial diseases

Helminthiasis

Human immunodeficiency virus [HIV] disease

Infections with a predominantly sexual mode of transmission

Intestinal infectious diseases

Mycoses

Other bacterial diseases

Other diseases caused by chlamydiae

Other infectious diseases

Other spirochaetal diseases

Other viral diseases

Cytomegaloviral disease

Infectious mononucleosis

Mumps

Other viral diseases, not elsewhere classified

Viral conjunctivitis

Viral infection of unspecified site

Adenovirus infection, unspecified

Coronavirus infection, unspecified

Enterovirus infection, unspecified

Other viral infections of unspecified site

Papovavirus infection, unspecified

Parvovirus infection, unspecified

Viral infection, unspecified

DetailsVisualizationNotes (0)Class Mappings (16)

Preferred Name

Coronavirus infection, unspecified

ID

<http://purl.bioontology.org/ontology/ICD10/B34.2>

cui

C0206750

notation

B34.2

prefLabel

Coronavirus infection, unspecified

tui

T047

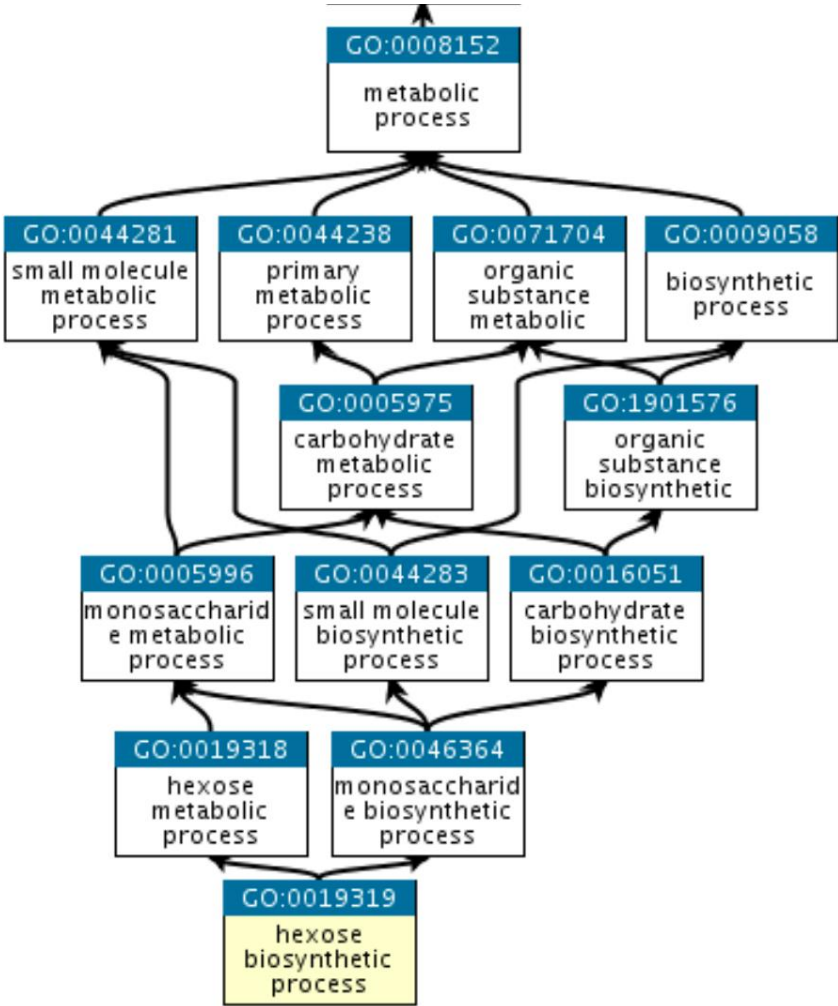
subClassOf

Viral infection of unspecified site

ICD-10 - Screenshot from

https://bioportal.bioontology.org/ontologies/ICD10/?p=classes&conceptid=http%3A%2F%2Fpurl.bioontology.org%2Fontology%2FICD10%2FB34.2&jump_to_name=true

For more about emergency use ICD codes for COVID-19 <https://www.who.int/standards/classifications/classification-of-diseases/emergency-use-icd-codes-for-covid-19-disease-outbreak>



Gene Ontology - Figure from

<https://geneontology.org/docs/ontology-documentation/>

Post-GPT: LM as Knowledge Base?

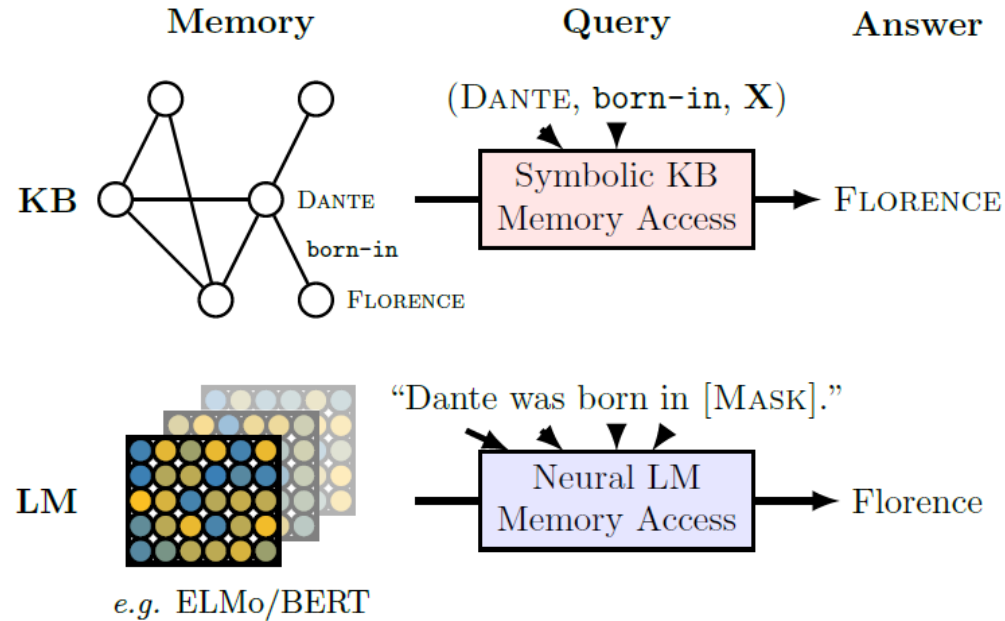


Figure 1: Querying knowledge bases (KB) and language models (LM) for factual knowledge.

Petroni F, Rocktäschel T, Riedel S, Lewis P, Bakhtin A, Wu Y, Miller A. *Language Models as Knowledge Bases?*. In EMNLP-IJCNLP, 2019 Nov.

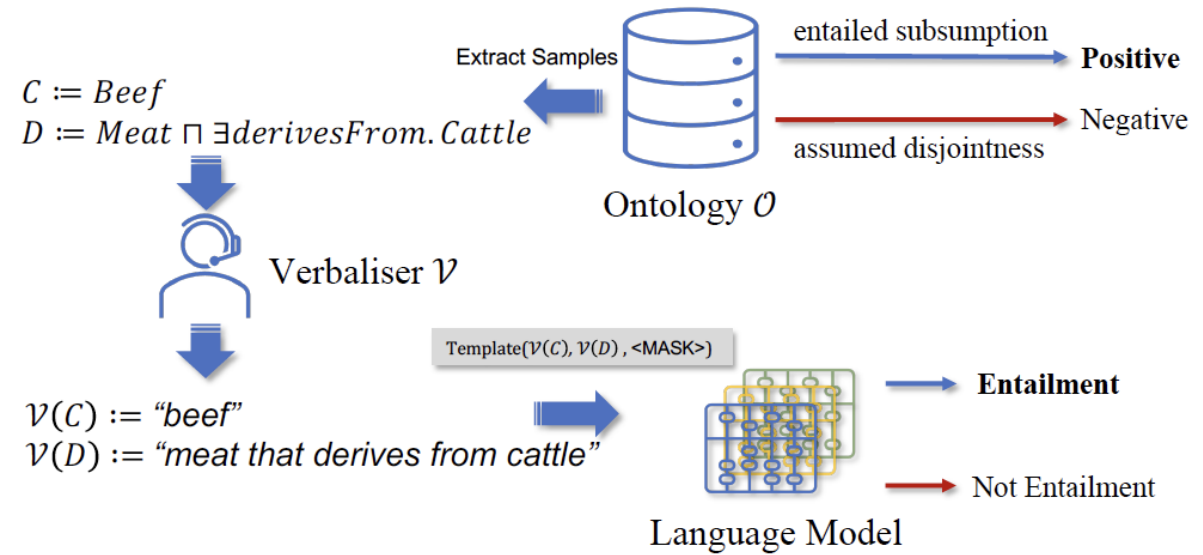


Figure 1: ONTO LAMA framework.

He Y, Chen J, Jimenez-Ruiz E, Dong H, Horrocks I. *Language Model Analysis for Ontology Subsumption Inference*. In Findings of the ACL, 2023 Jul.

LM and KG differences

- Distributional semantics
 - **Vectorised** representations
 - General & implicit data (“knowledge”)
 - Language processing
 - Data-driven
 - Neural network
- Formal semantics
 - **Structured** representations
 - Domain & explicit knowledge
 - Logical reasoning
 - Domain experts & data-driven
 - Symbolic / semantic

Three Pillars of “Machine Knowledge”

			
Representation:	crisp	mixed	<i>latent</i>
Correctness:	high	mixed	mixed
Freshness:	mostly	fresh	stale
Coverage:	limited	everything	huge
Cost/Query:	low	low	high
Provenance:	tangible	tangible	intangible
			
Coolness:	nerdy	mundane	amazing

Unifying KGs and LLMs

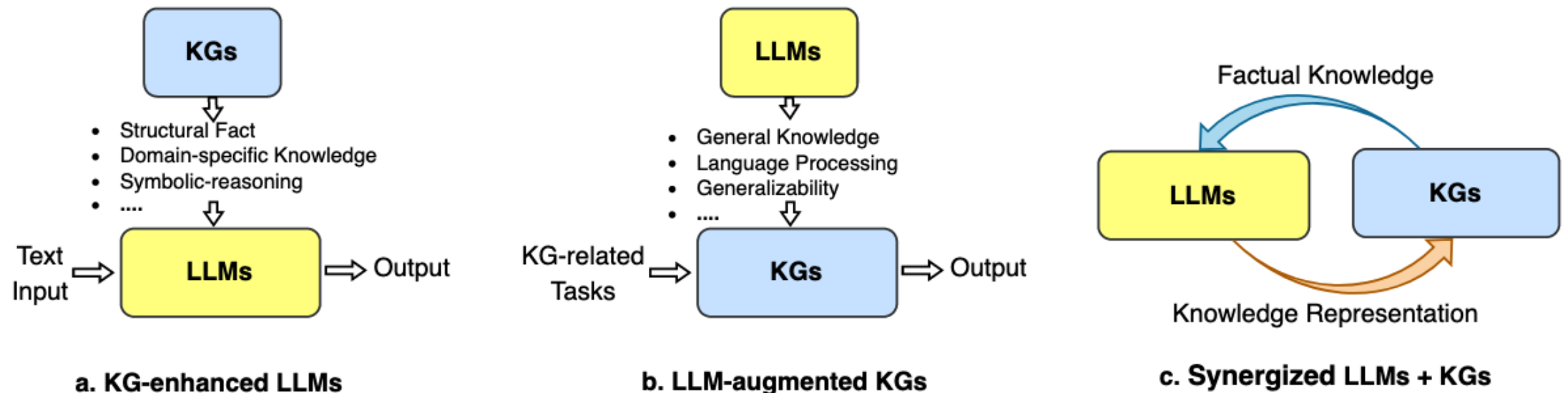


Fig. 6. The general roadmap of unifying KGs and LLMs. (a.) KG-enhanced LLMs. (b.) LLM-augmented KGs. (c.) Synergized LLMs + KGs.

Figure from Pan, S., Luo, L., Wang, Y., Chen, C., Wang, J., & Wu, X. (2023). Unifying Large Language Models and Knowledge Graphs: A Roadmap. *arXiv preprint arXiv:2306.08302*.

LLMs and KGs

- From Data to Application

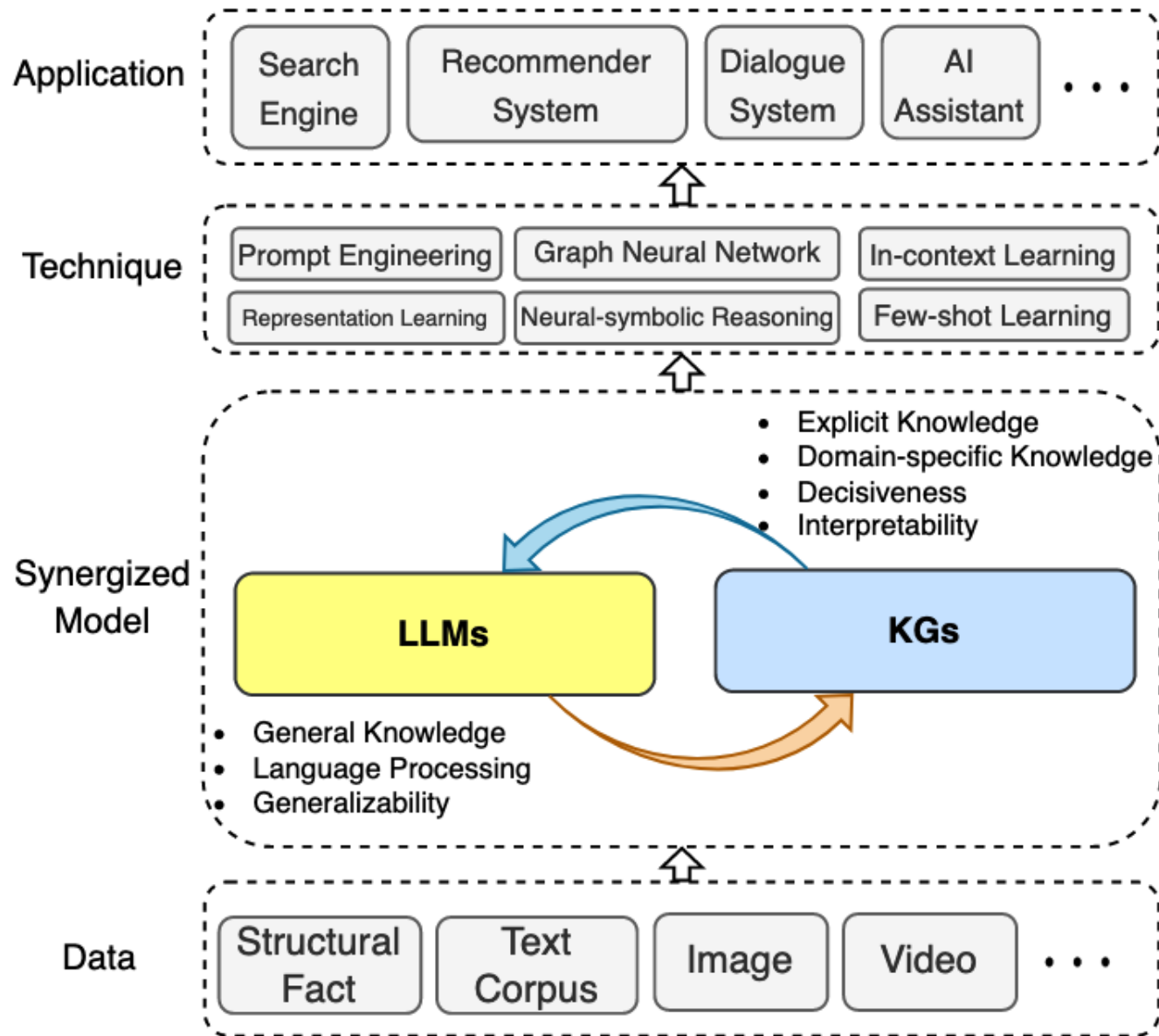


Figure from Pan, S., Luo, L., Wang, Y., Chen, C., Wang, J., & Wu, X. (2023). Unifying Large Language Models and Knowledge Graphs: A Roadmap. *arXiv preprint arXiv:2306.08302*.

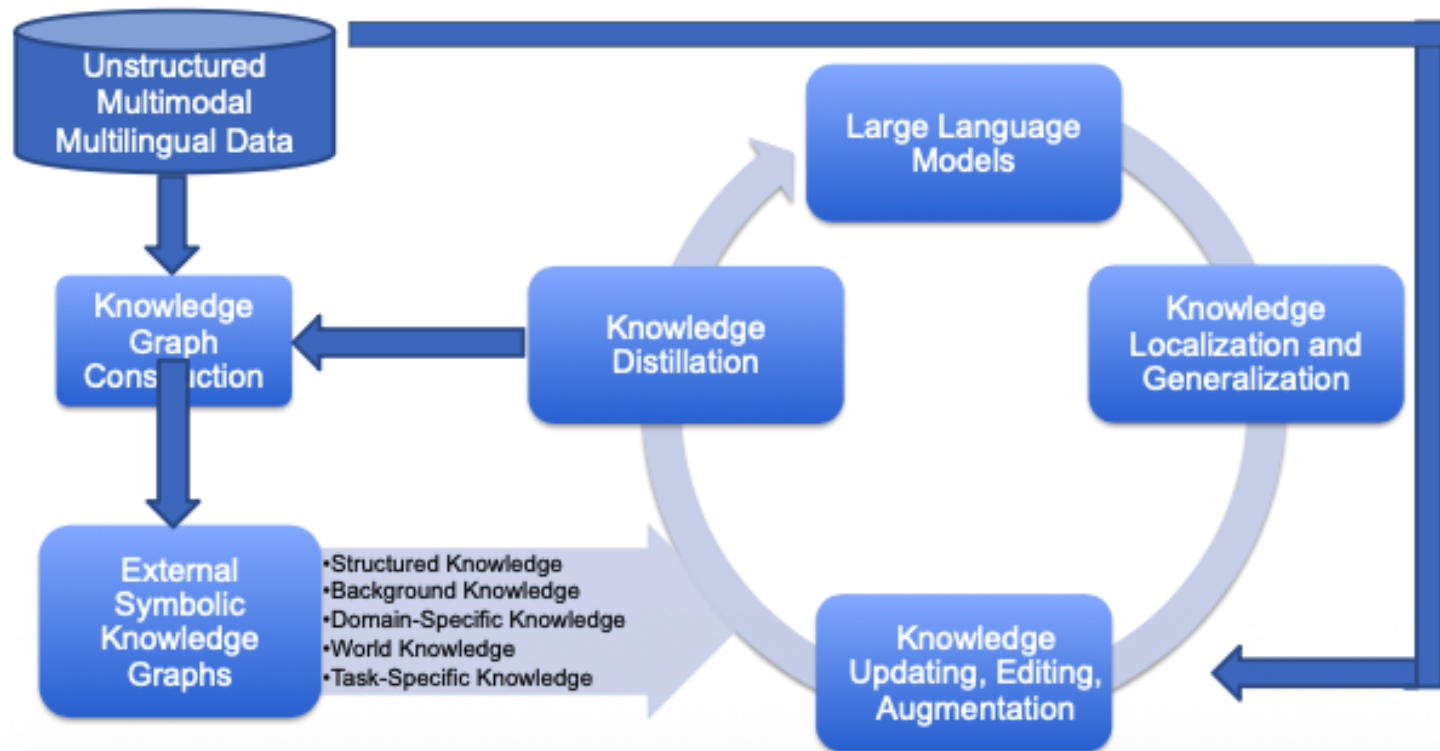
Fig. 7. The general framework of the *Synergized LLMs + KGs*, which contains four layers: 1) *Data*, 2) *Synergized Model*, 3) *Technique*, and 4) *Application*.

Lifecycle of Knowledge in LLMs

Figure from Manling Li, Zoey Sha Li, Yuji Zhang, Chi Han, Heng Ji. The Lifecycle of Knowledge in Large Language Models: Memorization, Editing, and Beyond. AAAI 2025 Tutorials. Feb 26, 2025.

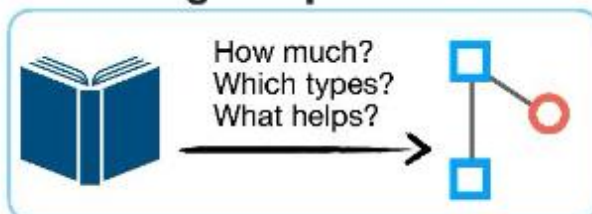
https://llmknowledge lifecycle.github.io/AAAI2025_Tutorial_LLMKnowledge/

Completing Knowledge Lifecycle by connecting LLMs and External Knowledge

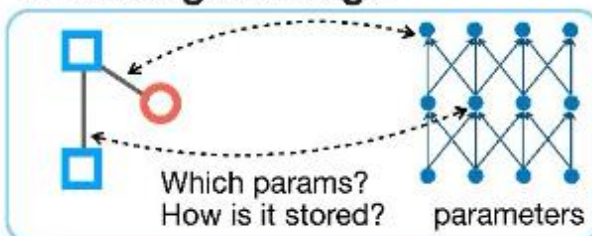


Tutorial Roadmap

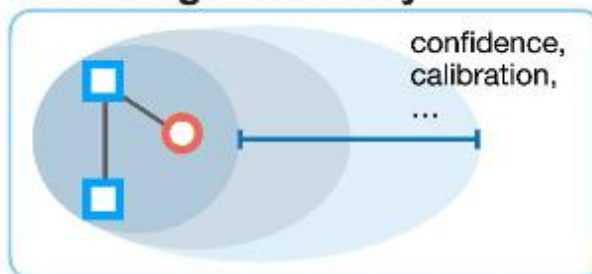
knowledge acquisition



knowledge storage

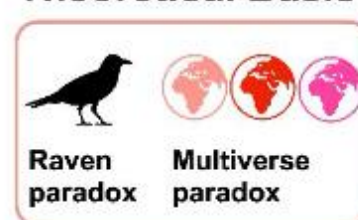


knowledge boundary

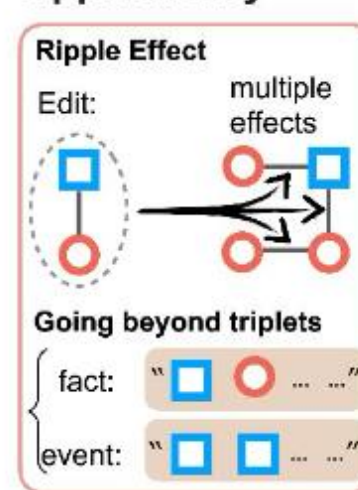


(b) Foundation: understanding LLMs' knowledge

Lack of A Theoretical Basis

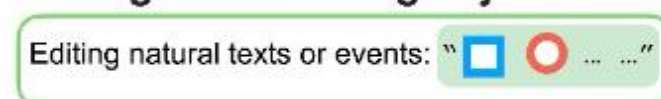


Limited Applicability

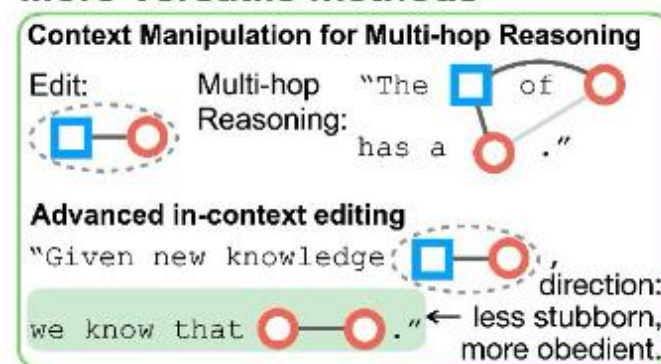


(c) Challenges in editing knowledge

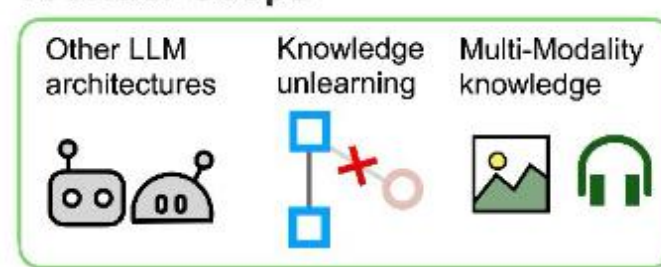
More general editing objective



More Versatile methods

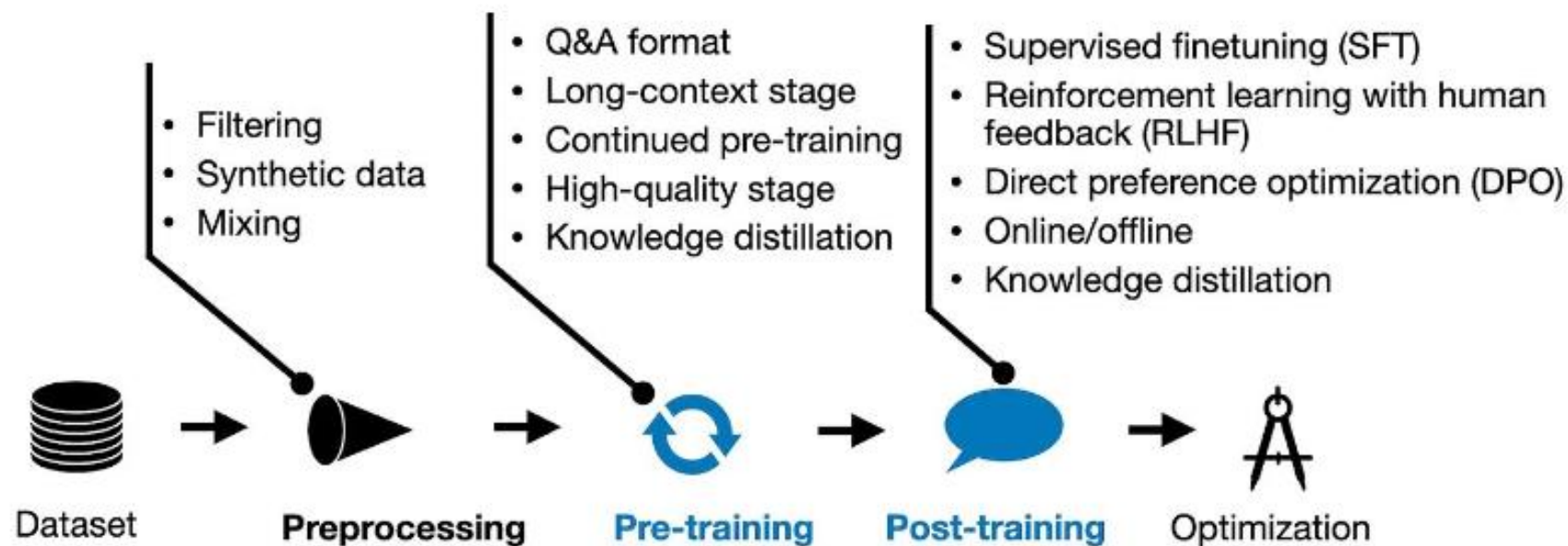


A Wider Scope



(d) Promising directions

Working with Frontier LLMs



- Knowledge is (mostly) acquired during pretraining
- Post-training for format and style alignment
- RAG systems and Agent systems to keep knowledge up-to-date

A knowledge perspective for medical imaging

- How do medical imaging AI use structured knowledge?
- What types of knowledge do they use?
- How explainability can be enabled by knowledge? (next week)



- Most knowledge used in previous work was in unstructured, tacit form.
- KG was highlighted in the future studies.

The knowledge graph

We believe that the knowledge graph (Wang et al., 2014), with the character of embedding different types of knowledge, is a generic and flexible approach to incorporate multi-modal medical domain knowledge. Although rarely used at present, it also shows advantage in medical image analysis tasks, especially in medical report generation (Li et al., 2019a). We believe that more types of knowledge graph can be used to represent and learn domain knowledge in medical image analysis tasks.

X. Xie, J. Niu, X. Liu, Z. Chen, S. Tang, and S. Yu, 'A survey on incorporating domain knowledge into deep learning for medical image analysis', *Medical Image Analysis*, vol. 69, p. 101985, Apr. 2021, doi: [10.1016/j.media.2021.101985](https://doi.org/10.1016/j.media.2021.101985).

Survey Paper

A survey on incorporating domain knowledge into deep learning for medical image analysis

Xiaozheng Xie^a, Jianwei Niu^{a,b}, Xuefeng Liu^{a,*}, Zhengsu Chen^a, Shaojie Tang^c, Shui Yu^d

^a State Key Laboratory of Virtual Reality Technology and Systems, School of Computer Science and Engineering, Beihang University, 37 Xueyuan Road, Haidian District, Beijing 100191, China

^b Beijing Advanced Innovation Center for Big Data and Brain Computing (BDBC) and Hangzhou Innovation Institute of Beihang University, 18 Chuanghui Street, Binjiang District, Hangzhou 310000, China

^c Jindal School of Management, The University of Texas at Dallas, 800 W Campbell Rd, Richardson, TX 75080-3021, USA

^d School of Computer Science, University of Technology Sydney, 15 Broadway, Ultimo NSW 2007, Australia

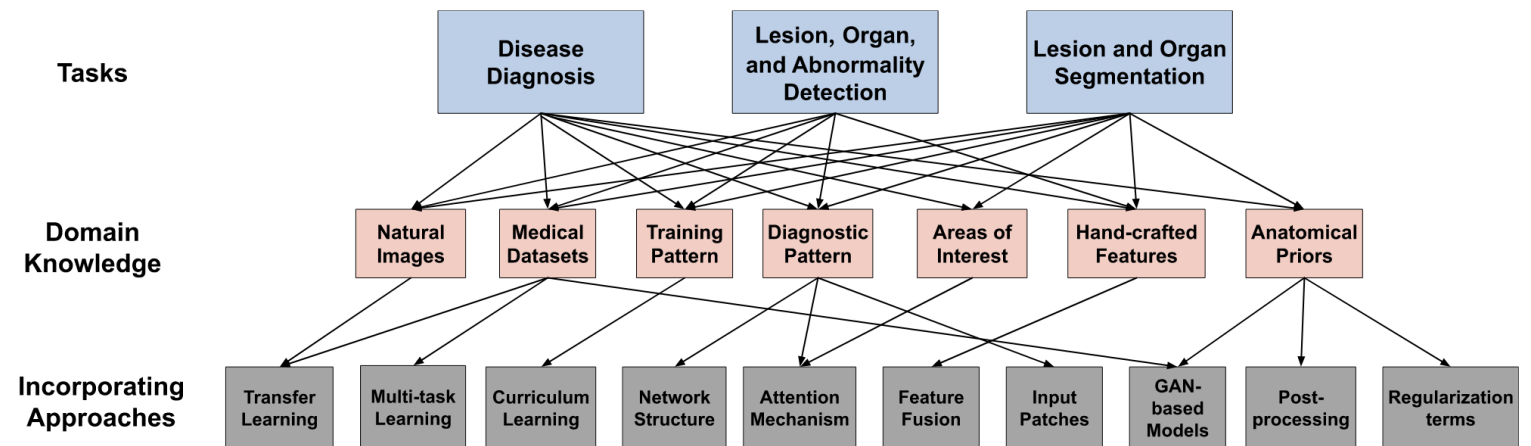
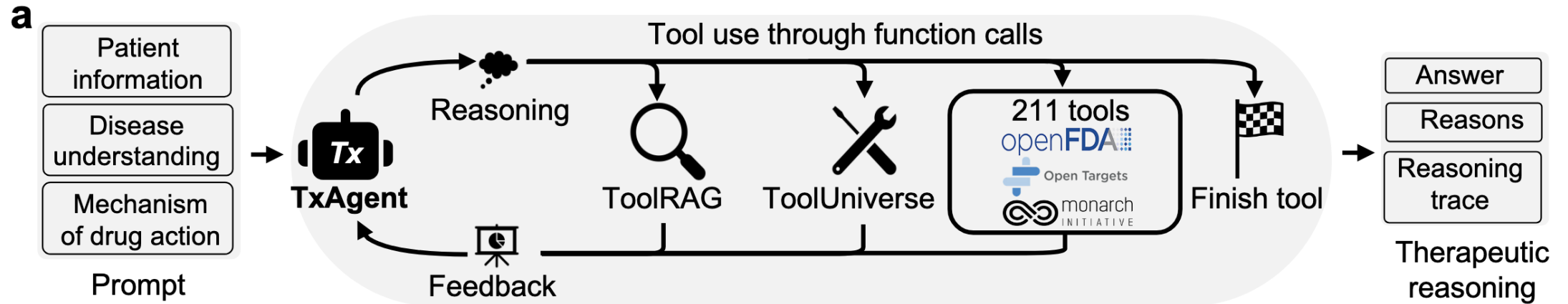


Fig. 1. Methods of information categorization and incorporating methods in disease diagnosis; lesion, organ, and abnormality detection; lesion and organ segmentation.

Structured knowledge used in MICCAI 2025

- Knowledge graphs as tools in agentic AI
- Multimodal hierarchical structure of images and scene graphs
 - Concept-guided graphs
 - Multimodal hierarchies & concept hierarchies
 - Scene graphs
- Keywords or concept lists
- LLM-generated structured knowledge
 - concept-attribute list
 - text or descriptions

Knowledge graphs as tools in agentic AI

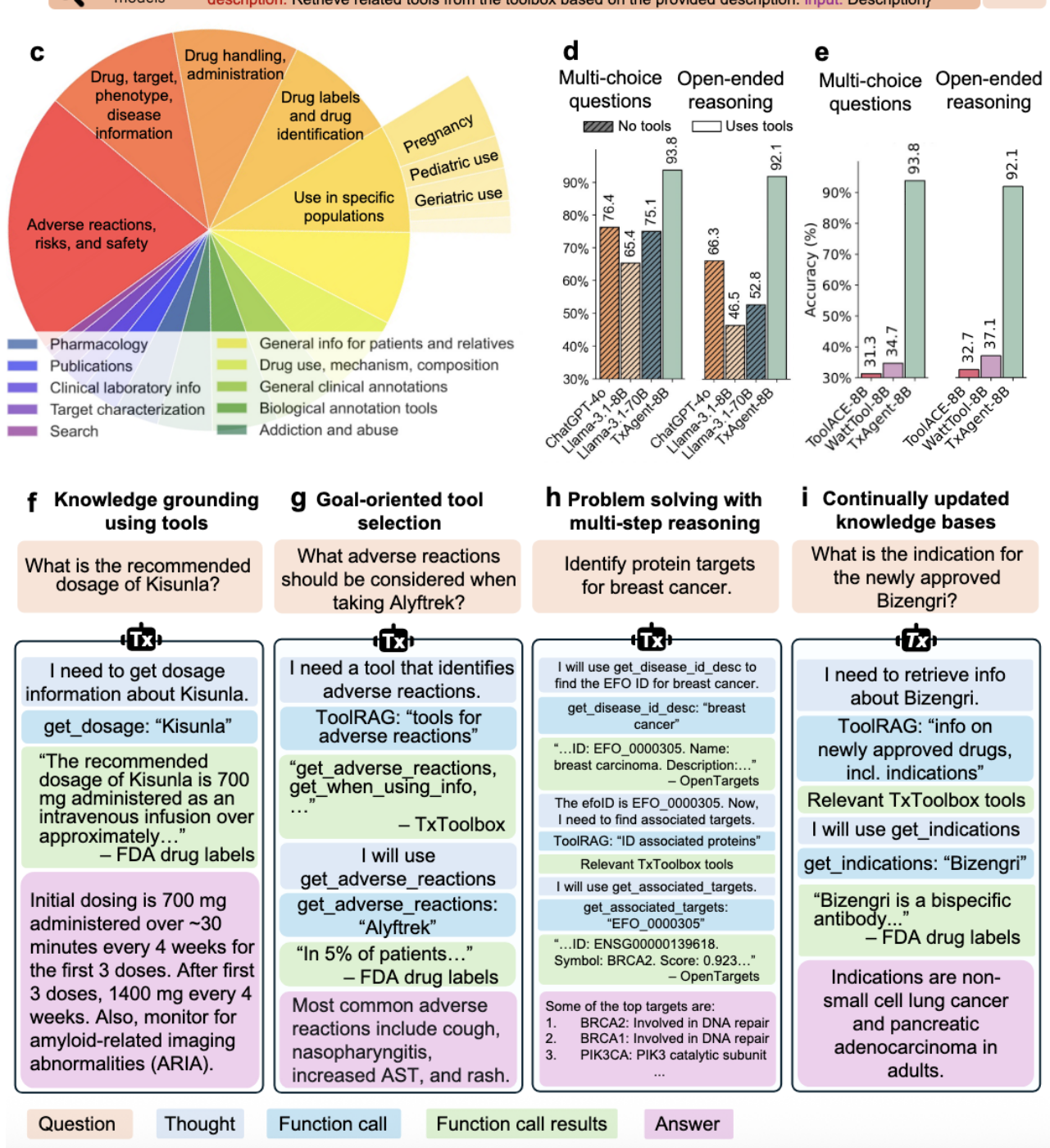


b

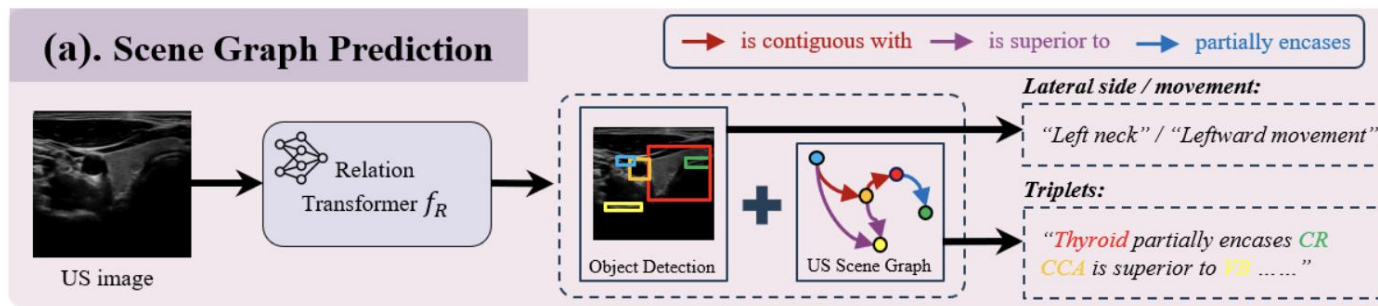
	{name: get_associated_targets_by_disease_efoID description: Find targets associated with a specific disease or phenotype based on EFO ID. input: EFO ID}	52 tools
	{name: get_contraindications_by_drug_name description: Retrieve contraindications information based on the drug name. input: Drug name}	156 tools
	{name: get_associated_diseases_by_HPO_ID description: Retrieve diseases associated with a list of phenotypes by the phenotype HPO IDs. input: HPO ID}	3 tools
	{name: Tool_RAG description: Retrieve related tools from the toolbox based on the provided description. input: Description}	1 tool

S. Gao *et al.*, 'TxAgent: An AI Agent for Therapeutic Reasoning Across a Universe of Tools', Mar. 14, 2025, *arXiv*: arXiv:2503.10970. doi: [10.48550/arXiv.2503.10970](https://doi.org/10.48550/arXiv.2503.10970).
<https://zitniklab.hms.harvard.edu/TxAgent/>

S. Gao *et al.*, ‘TxAgent: An AI Agent for Therapeutic Reasoning Across a Universe of Tools’, Mar. 14, 2025, *arXiv*: arXiv:2503.10970. doi: [10.48550/arXiv.2503.10970](https://doi.org/10.48550/arXiv.2503.10970).

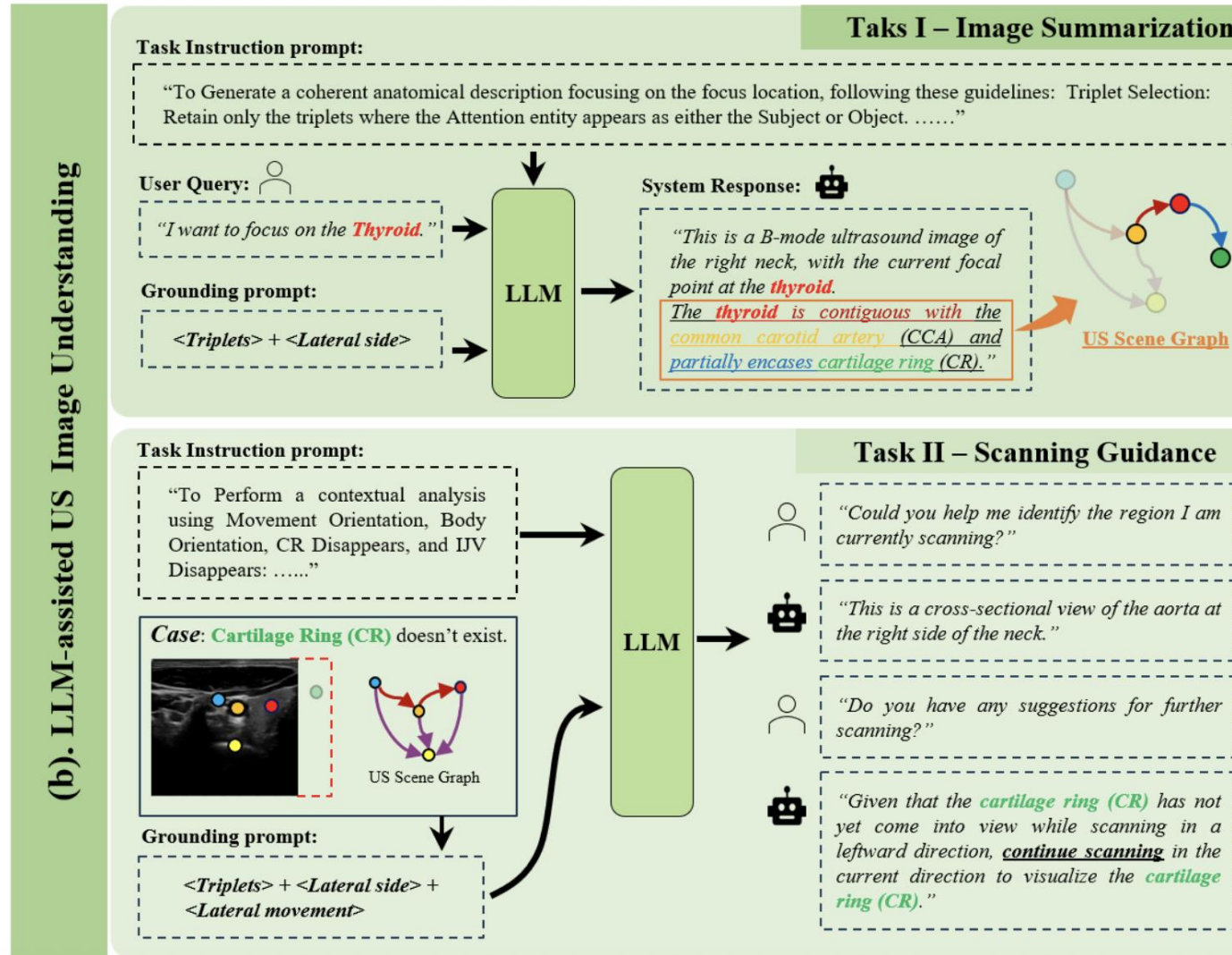


Scene graphs



Semantic Scene Graph for
 Ultrasound Image
 Explanation and Scanning
 Guidance

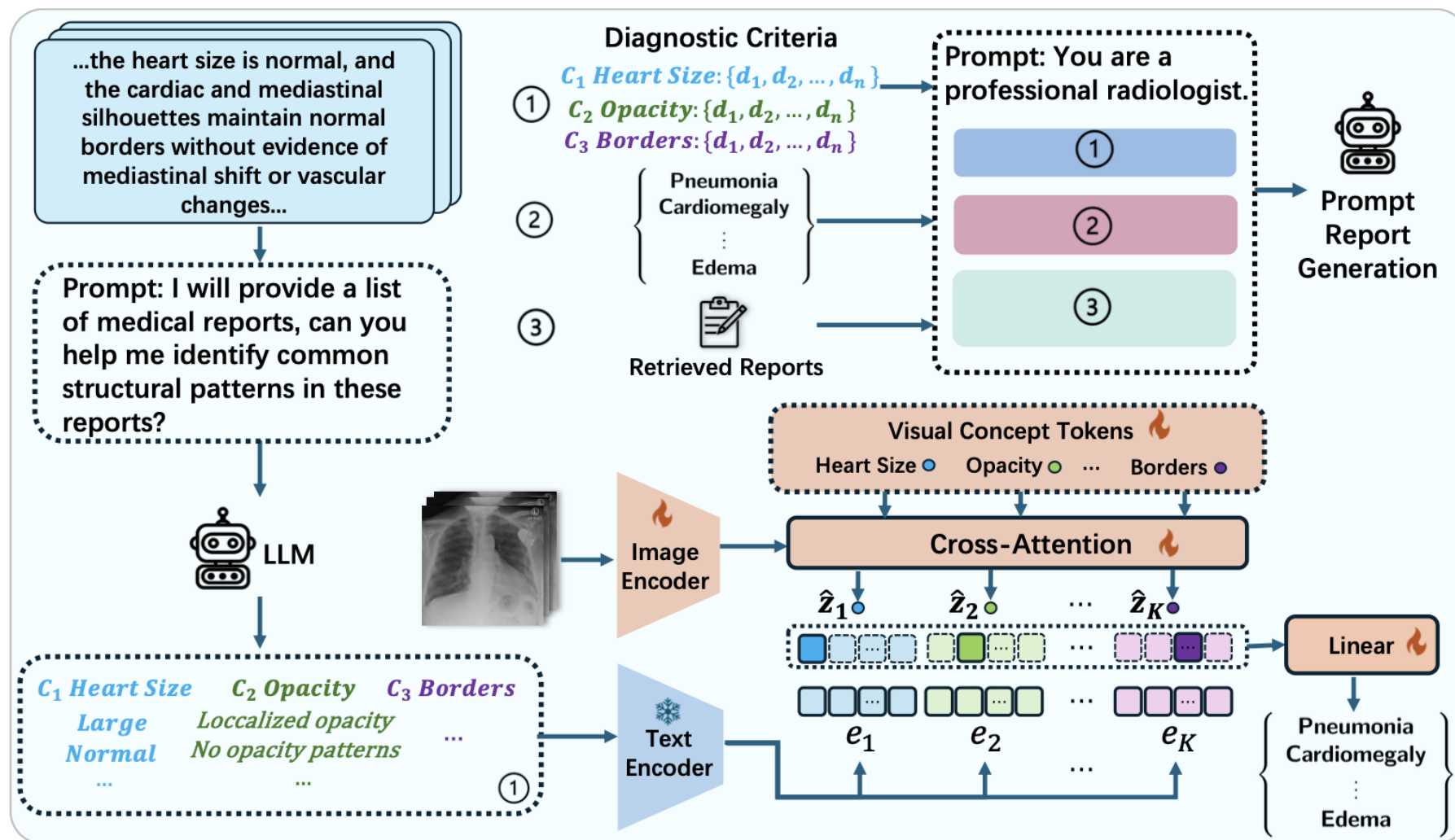
<https://papers.miccai.org/miccai-2025/0819-Paper3129.html>



LLM generated knowledge

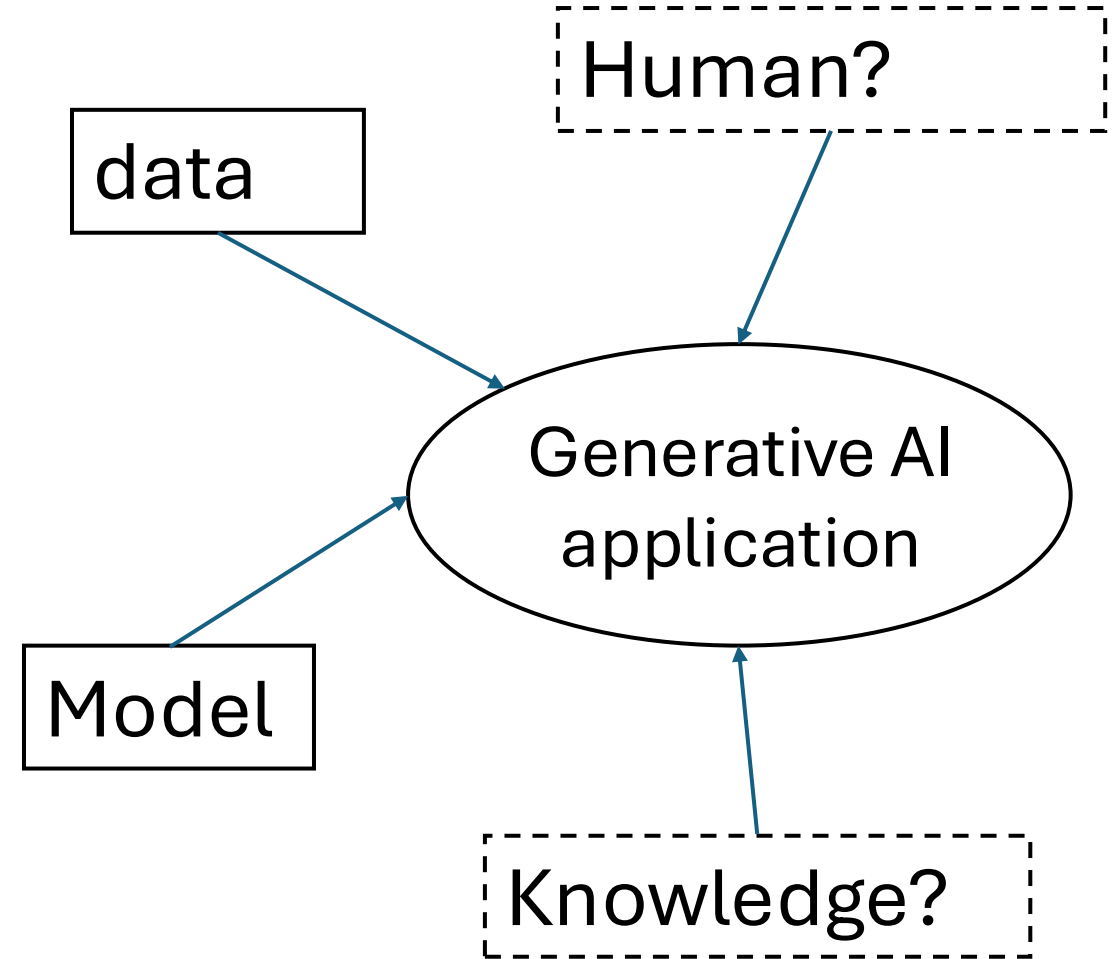
RadAlign: Advancing
Radiology Report Generation
with Vision-Language
Concept Alignment

<https://papers.miccai.org/miccai-2025/0741-Paper3308.html>



Discussion

- What is your GenAI application?
- What is data? How to represent the data? Additional knowledge you would need?
- What models do you use?

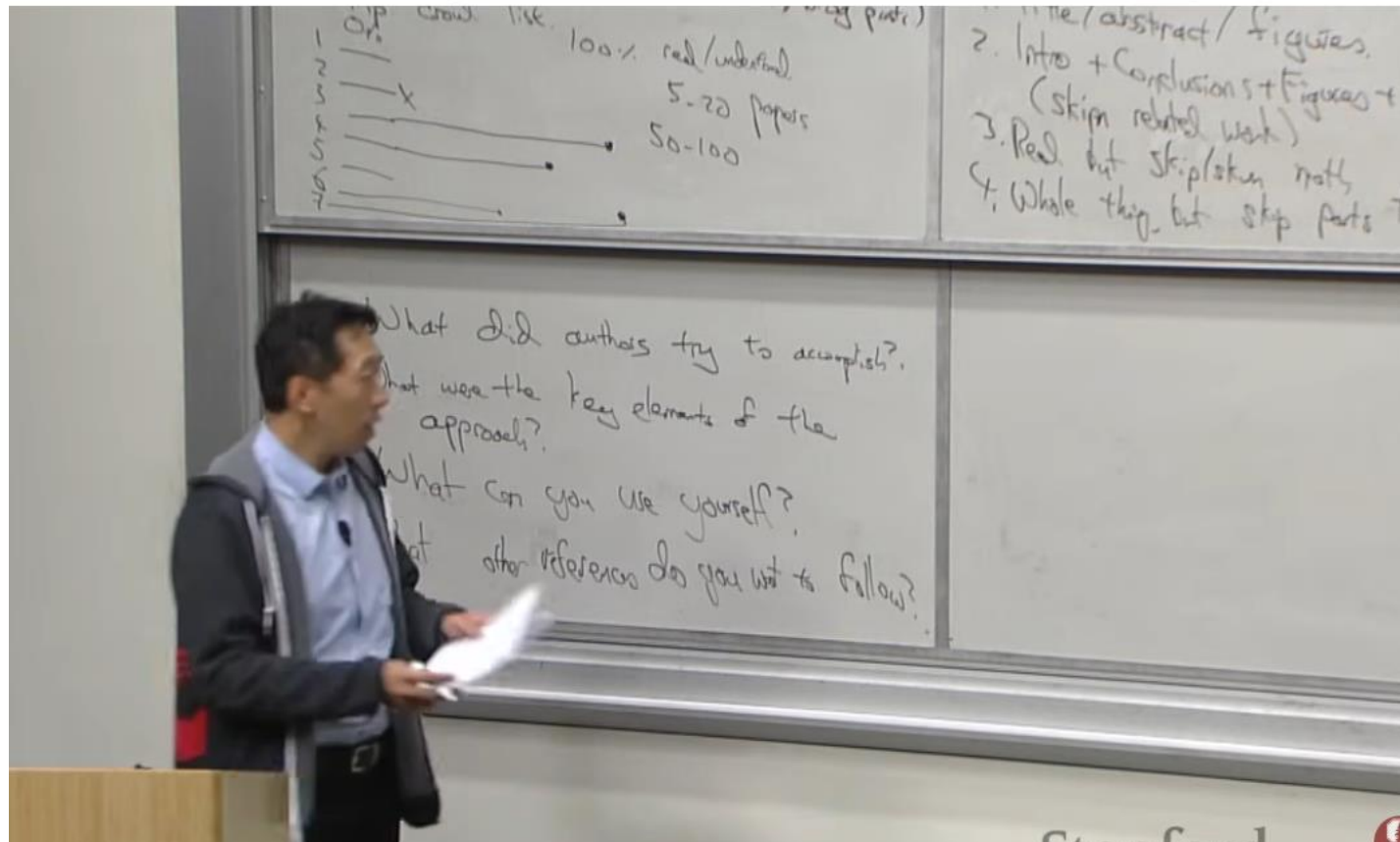


Conclusion

- Knowledge can be both structured (KG) and parametric (LLM) in GenAI applications: they can enhance to each other and synergise.
- The idea of knowledge provides us way to understand LLMs, e.g., knowledge update, unlearning, boundary / uncertainty, distillation, explanation, etc.
- We can start to plan our GenAI applications.

How to read a paper?

- <https://www.youtube.com/watch?v=733m6qBH-jl> – about how to read a paper “Stanford CS230: Deep Learning | Autumn 2018 | Lecture 8 - Career Advice / Reading Research Papers”



Friday 6 Feb 2026 – paper discussion

- A. Shmatko *et al.*, ‘Learning the natural history of human disease with generative transformers’, *Nature*, pp. 1–9, Sep. 2025, doi: [10.1038/s41586-025-09529-3](https://doi.org/10.1038/s41586-025-09529-3).
- Code for Delphi and accompanying scripts and Jupyter notebooks are available at GitHub (<https://github.com/gerstung-lab/delphi>).
- Supplementary information https://static-content.springer.com/esm/art%3A10.1038%2Fs41586-025-09529-3/MediaObjects/41586_2025_9529_MOESM1_ESM.pdf

nature

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Article | [Open access](#) | Published: 17 September 2025

Learning the natural history of human disease with generative transformers

[Artem Shmatko](#), [Alexander Wolfgang Jung](#), [Kumar Gaurav](#), [Søren Brunak](#), [Laust Hvas Mortensen](#), [Ewan Birney](#) , [Tom Fitzgerald](#)  & [Moritz Gerstung](#) 

Nature **647**, 248–256 (2025) | [Cite this article](#)

265k Accesses | 23 Citations | 1692 Altmetric | [Metrics](#)

An [Author Correction](#)



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articles/s41586-025-09529-3#Sec51

Learning the natural history of human disease with generative transformers

Supplementary information

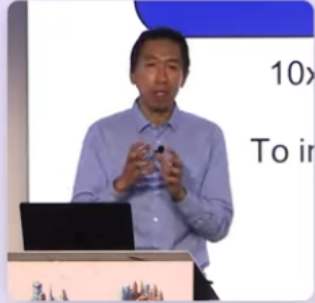
[Supplementary Information](#)

Supplementary Figs. 1–12, Supplementary Table 1 and Supplementary Discussion.

Prototyping with AI

**AI Dev 25
x NYC**

DeepLearning.AI



10x
To in

Building with AI coding

**Build
Prototypes**

10x faster!

**Write/maintain
production
software**

50% faster?

To innovate, build 20 prototypes to see what works

~~Move fast and break things.~~
Move fast and be responsible.

Andrew Ng

<https://www.youtube.com/watch?v=6ejKX20es3o> Deeplearning.ai.

AI Dev 25 x NYC | Andrew Ng: Opening Keynote. 2 Dec 2025.

Wednesday 4 Feb 2026

- Tasks (choose one below):
 - Training a small LLM with NanoGPT <https://github.com/karpathy/nanoGPT>
 - Testing the knowledge of an LLM, e.g., using the LAMA probe <https://github.com/facebookresearch/LAMA>
- Prototyping of your project
 - Q&A
 - Test feasibility with prototyping
 - Looking at the literature
 - AI-supportive